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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

PhD Prospectus

**MULTI-LAYER SECURITY FOR MOBILE USERS:
MEASUREMENT AND MITIGATION FROM APPLICATIONS TO CONVERSATION**

By

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Abstract

Mobile users can lose control of their data and money through harmful applications, platform mechanisms that obscure changes in access, and deceptive conversations. This dissertation investigates these three settings through measurement and the assessment of practical defenses. Two studies provide its foundation. *The Cost of Convenience* examines loan applications across five countries and connects policy requirements to observable application behavior; coordinated disclosure led to the removal of 93 applications from Google Play. *Silent Consent, Persistent Risk* studies permission expansion and cross-application access in Android, combining a longitudinal corpus of 19.3 million APKs with targeted device experiments and a transparency prototype. The ongoing third study, *SIRENBERT*, investigates whether a sequence of observable conversational behaviors can support timely warnings about financial scams on the user's device. An encoder-recurrent-model prototype and a proxy conversation corpus provide a starting point, while annotation validation, generalization, warning timing, and deployment evaluation remain research tasks. The proposed extension examines images and other media presented as evidence of identity, circumstances, or financial opportunity. I will reconstruct conversations and embedded images from public screenshot collections, measure how these artifacts are used, and test whether their interpretation in context improves warnings over text alone. The central question is how evidence available at each layer can be translated into useful protection while preserving user privacy and controlling false alarms. The dissertation will distinguish demonstrated findings from the hypotheses that these remaining studies will test.

1 Introduction

Mobile devices mediate access to financial services, private communication, and personal information. Their importance is especially visible where mobile money provides access to services that conventional banking does not readily reach [1, 2]. This dependence creates several opportunities for abuse. An application may collect information inconsistent with its stated purpose; a platform may grant access without making the change visible; and a person using an otherwise legitimate messaging application may be persuaded to send money under false pretenses. These problems involve different evidence and intervention points, but share a concern: users may have little opportunity to recognize a developing risk before a consequential action occurs.

My dissertation examines this concern at the application, platform, and conversation levels. The first study measures how loan applications comply with regulatory and platform requirements and how their runtime behavior exposes sensitive data [3]. The second investigates Android permission mechanisms that allow capabilities or cross-application access to expand without renewed user visibility [4]. Together, these studies establish methods for connecting policy or permission abstractions to observable behavior and for assessing interventions in realistic settings.

The remaining research addresses conversational financial exploitation. Access controls can constrain an application’s use of data, but they do not determine whether a request made through that application is trustworthy. A request for money can be legitimate, and behaviors such as affection, professional self-presentation, and urgency also occur outside scams. The research problem is therefore to determine whether the evolving context of an exchange provides useful evidence for a timely warning, without treating ordinary persuasion or financial discussion as sufficient evidence of fraud.

Thesis proposition. *Measuring how access and trust develop over time can reveal risks that are obscured by isolated application, permission, or message observations. These measurements can inform defenses at the point where the relevant evidence is available, with their value assessed through detection quality, timeliness, user visibility, and deployment cost.*

This proposition does not require a single detector for all three settings. The completed studies provide evidence at the application and platform levels. The ongoing conversation study tests behavioral sequence modeling, and the proposed extension asks whether images presented within an exchange add useful evidence. Sections 3 and 4 summarize completed work; Sections 5 and 6 specify what remains to be established.

2 Background and Research Gaps

2.1 Application behavior and permission visibility

Digital lending can improve access to credit, but loan applications also create opportunities for intrusive data collection and coercive debt recovery. Registration requirements and platform permission policies provide complementary constraints. The application-level question is whether the behavior of approved applications is consistent with those requirements, rather than whether approval alone indicates safety. My loan-app study connects policy interpretation to static and dynamic measurements [3].

Android permissions have been studied in terms of application privileges and users’ understanding of access requests. Felt et al. examined permission use and user attention, while Wijesekera et al. investigated the context in which users make permission decisions [5–7]. Calciati et al. measured automatically granted permissions across application versions; Tuncay et al. studied custom-permission weaknesses; and Gamba et al. examined preinstalled software [8–10]. My platform study builds on this work by measuring permission evolution at larger scale and testing selected access paths on Android 16. Its prototype addresses visibility into changes, rather than claiming to prevent all misuse of an otherwise granted permission.

2.2 Conversational scams and evidence beyond text

Whitty describes the persuasive progression of romance fraud. Oak and Shafiq use victim interviews to examine relationship-based investment scams, including trust development, financial exploitation, and subsequent re-engagement. Acharya and Holz investigate the practices and infrastructure associated with pig-butchering scams [11–13]. These studies motivate examining connected behaviors over time; they do not imply that every scam follows one fixed sequence or that a particular tactic uniquely identifies fraud.

Recent detection work provides useful comparison points. Lu et al. study mobile messaging scams using Reddit data, Sun et al. introduce a benchmark for scam progression, and Topcuoglu et al. compare generative-model approaches to scam detection [14–16]. Their settings and data sources motivate complementary questions about message-level evidence, transfer between sources, and device constraints. Our scam-family labels will describe our own data; an external benchmark’s categories are supporting context rather than a required taxonomy.

The remaining gap is the reliability and utility of a representation that can be updated as a private conversation unfolds. This requires appropriate negative examples, validated message labels, and evaluation before the financial request. It also raises a multimodal question: a photograph, document, or balance screenshot may be presented as proof, while its relevance to a later request depends on what has already been said. Detecting image reuse or manipulation alone does not establish that an exchange is exploitative. The proposed work studies the artifact’s role and contextual interpretation together.

3 Completed Work I: The Cost of Convenience

The Cost of Convenience: Identifying, Analyzing, and Mitigating Predatory Loan Applications on Android, published at ACM ASIA CCS 2026, investigates whether loan applications comply with national rules and Google’s Financial Services Policy [3].

Problem and approach. An approved lender can still operate an application that collects prohibited data or exposes it before a borrower has completed registration. We studied 434 applications from registries and markets in Indonesia, Kenya, Nigeria, Pakistan, and the Philippines. An LLM-assisted policy-to-permission mapping translated policy text into testable checks, which we combined with static code inspection and dynamic observation of network and data-access behavior.

Findings and contribution. Among approved applications, 141 violated national regulatory requirements and 147 violated platform policy. In the dynamic analysis, 37 of 148 functioning applications transmitted sensitive information at launch after permissions were granted, before signup or a loan application. Following coordinated disclosure, Google removed 93 flagged applications, representing over 300 million cumulative installs. These installation counts indicate potential exposure, not a count of verified victims. The study contributes a measurement method and evidence for more consistent compliance assessment across regulatory and platform requirements.

4 Completed Work II: Silent Consent, Persistent Risk

Silent Consent, Persistent Risk: Android Permission Groups and Custom Permissions examines how application capabilities and data access can change without corresponding user visibility [4].

Problem and approach. A permission decision made for one application version may not convey what a later version can access. Custom permissions can also protect components in ways that admit unrelated applications. We analyzed 19,324,760 APKs representing 5,965,583 applications in AndroZoo [17], tracked permission changes, and compared their association with VirusTotal detections across thresholds [18]. Targeted Android 16 experiments tested whether selected mechanisms remained exploitable. A prototype using

public Android APIs exposed otherwise silent permission changes to the user.

Findings and limits. Of 2,244,575 multi-version applications, 381,026 (17%) exhibited within-group permission expansion. Malware-flagged applications had higher odds of expansion at the primary threshold (odds ratio 1.35), an association rather than evidence that every expansion is malicious. Static analysis identified 307 active cross-developer access pairs. Dynamic retrieval of sensitive data was confirmed for the 39 pairs amenable to probing on Android 16; the dynamic claim does not extend to all 307 pairs. In a 96-day single-device pilot, the transparency prototype reported 23 expansion events across 13 applications. This establishes feasibility of providing visibility, while broader usability and protective effects require further evaluation.

5 Ongoing Work III: Conversational Scam Detection

SIRENBERT investigates whether observable behaviors, interpreted in sequence, can support early warnings about conversational scams on a mobile device. A text-classification and recurrent-model prototype has been implemented, including an Android messaging-client integration. Establishing the reliability of its supervision, its ability to generalize, and the usefulness of its warnings remains part of this dissertation.

5.1 Problem, scope, and working representation

A *conversational scam* is a multi-turn exchange in which a suspect uses false pretenses to induce a target to part with money or other economic value through a request or instruction. The three defining aspects are a multi-turn interaction, deception, and a financial demand. The study includes romance, investment, advance-fee, employment, and related financial scams where these aspects are present. Our evaluated scope focuses on suspect-requested financial actions; coercion-based sextortion is held separately.

This problem definition is broader than what a transcript can establish. We do not independently verify an occupation, emergency, or promised return. For annotation, we identify observable behavior and assess whether an explicit financial demand is supported as exploitative by its preceding context. A demand provides a reproducible reference point for studying a scam attempt; no completed payment is required, and the demand need not end the transcript. Earlier concerning behavior can warrant an early-warning hypothesis without being labeled a confirmed scam attempt. A direct request for money, by itself, is insufficient to establish exploitation.

The current vocabulary contains fourteen triggers (Table 1). They describe conversational functions rather than scam families or proven intent. For example, wealth and professional standing can both supply a claim to credibility, motivating their initial grouping. We will test that grouping through annotation reliability and finer-granularity comparisons. A common state transition is not sufficient proof that distinctions are unnecessary, and assigning every message a label does not establish complete behavioral coverage. Independent review will identify missed known acts, overlaps hidden by single-label precedence, and relevant acts absent from the vocabulary.

Let a_t denote the trigger for suspect message t , and let the ordered state $s_t \in \{0, 1, 2, 3, 4\}$ represent no observed behavior, initial contact, relationship building, pre-exploitation behavior, and a demand after setup. Define $b(a)$ as 1 for contact/maintenance, 2 for relationship behaviors, and 3 for the eight setup behaviors. The current deterministic summary is

$$s_0 = 0, \quad s_t = \begin{cases} 4, & a_t = D \text{ and } s_{t-1} \geq 3, \\ \max(s_{t-1}, 1), & a_t = D \text{ and } s_{t-1} < 3, \\ \max(s_{t-1}, b(a_t)), & a_t \neq D, \end{cases} \quad (1)$$

Trigger	Observable behavior	State floor
WrongNumber_Intro	Mistaken-contact framing with visible continued engagement	Contact
Introduction_Opener	Initial greeting or introduction	Contact
SmallTalk_Maintenance	Routine discussion, logistics, or acknowledgment	Contact
Trust_Building	Personal disclosure or invitation to confide	Relationship
Affectionate_Language	Warmth without explicit romantic commitment	Relationship
Credential_Positioning	Standing, resources, or expertise presented for credibility	Setup
Gain_Offer	Prospective financial benefit offered to the target	Setup
Verification_Avoidance	Repeated checking deflection or suspect-controlled proof	Setup
Platform_Migration	Proposal to continue on another communication channel	Setup
Love_Bombing	Explicit romantic declaration or commitment	Setup
Crisis_Disclosure	Acute personal emergency presented	Setup
Emotional_Manipulation	Guilt, obligation, or relational leverage	Setup
Urgency_Pressure	Pressure for prompt action	Setup
Financial_Demand	Request/instruction that would cost the target economic value	Guarded

Table 1: Working behavioral vocabulary. These acts also occur in legitimate exchanges. State floors govern bookkeeping, not the legitimacy of a message. The legacy name Love_Bombing denotes the operational declaration test here.

where D denotes Financial_Demand. Thus ordinary messages do not erase previously observed setup, and setup need not be preceded by a rapport phase (Figure 1). This is a working summary of observed behavior, not a claim that every person follows a monotonic psychological process.

Conversation labels are derived separately. Let $q_t = 1$ only when a demand is interpreted as exploitative and its cited earlier setup messages support that interpretation. Citations must resolve to the correct preceding messages; human review must assess their semantic support. For an eligible, reviewed conversation C , the operational rule is

$$Y(C) = \begin{cases} \text{SCAM}, & \exists t : a_t = D \wedge q_t = 1, \\ \text{SUSPICIOUS}, & \text{otherwise, if a setup trigger occurs,} \\ \text{NON-SCAM}, & \text{otherwise.} \end{cases} \quad (2)$$

Unresolved annotations will be withheld from conversation-level supervision rather than converted into negative targets. The state and label can disagree: a legitimate payment request after a crisis can reach state 4 without satisfying Equation 2’s scam condition.

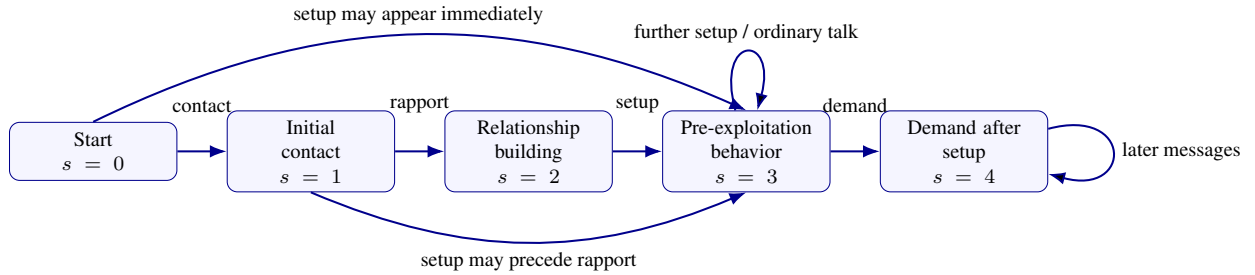


Figure 1: Illustrative transitions in the working lifecycle summary. All states can persist, and observed conversations may skip stages. Reaching the final state does not itself establish exploitation.

5.2 Proxy dataset and preliminary observations

Direct, complete records of private scam conversations are scarce. We therefore assembled a proxy corpus combining recorded exchanges, public screenshots, ordinary dialogue, and legitimate persuasive requests. LoveFraud02 contains researcher-recorded exchanges with purported romance-fraud perpetrators; Reddit Scambait provides screenshot-based exchanges involving people who knowingly engage suspected scammers [14, 19]. Neither source should be described as a representative sample of unsuspecting victims. Missing screenshots, selective posting, performative responses, and uncertain role attribution limit what they establish.

Topical-Chat and a DailyDialog-derived subset provide ordinary dialogue, while PersuasionForGood provides financially consequential but legitimate donation requests [20–22]. The latter is particularly important for distinguishing persuasion from fraud. Generated investment/relationship conversations and report-grounded ScamForge conversations extend the available behavior combinations. Their intended scam or benign status is a generation condition, not a verified label. They will support controlled stress tests and augmentation experiments, with primary real-source evaluation reported separately.

Preliminary result. In the current rule-derived analysis, 520 conversations from the real scam-source collections receive a scam label. They contain a median of two distinct mapped indicators before the first financial demand; 52.3% contain at least two. Among 1,685 benign-source conversations containing a demand but not receiving a scam label, 90.1% contain no mapped indicator before that demand. These observations motivate studying context, but are not independent evidence of detector accuracy: both the indicators and cohort labels are derived from provisional annotations. The pilot will examine label errors and legitimate uses of the same behaviors.

The new Reddit collection will expand candidate coverage, but its posts and images will enter this table only after conversation reconstruction, deduplication, and quality review. Increasing the number of screenshots alone does not resolve the corpus’s source bias.

5.3 Detection model and remaining evaluation

Stage 1: interpreting the message. Let m_t be the current suspect message and $\mathcal{H}_t^{(k)}$ the preceding context, including both speakers. A message classifier produces a distribution over the fourteen behaviors:

$$\mathbf{z}_t = \text{softmax}\left(f_{\theta}(m_t, \mathcal{H}_t^{(k)})\right), \quad \mathbf{z}_t \in [0, 1]^{14}. \quad (3)$$

Source	Conversations	Annotated suspect messages
LoveFraud02	83	26,423
Reddit Scambait	1,142	64,290
Topical-Chat	8,628	97,204
DailyDialog-derived subset	3,100	19,981
PersuasionForGood	1,017	10,600
Generated pig-butcherer scenarios	5,683	237,296
Generated ScamForge scenarios	3,097	152,371
Total	22,750	608,165

Table 2: Existing proxy corpus after the documented scope exclusions, before adding the new Reddit crawl. Message counts describe annotated suspect-side messages, not all messages exchanged. Source provenance is distinct from the operational labels; revised annotations remain subject to validation.

The prototype uses compact Transformer encoders. I will compare DistilBERT, ModernBERT, and an on-device LLM adapted for behavioral classification, including appropriate distilled or quantized variants [23–25]. A common behavioral interface will support comparison of contextual accuracy, latency, memory, and energy. A larger context window or model is an experimental choice, not an assumed improvement.

Stage 2: interpreting the sequence. The prototype passes trigger probabilities to a unidirectional GRU [26]:

$$\mathbf{h}_t = \text{GRU}_\phi(\mathbf{z}_t, \mathbf{h}_{t-1}), \quad r_t = \sigma(\mathbf{w}^\top \mathbf{h}_t + c). \quad (4)$$

Here $\mathbf{h}_t$ retains information across messages and r_t is a learned risk score. The learned memory is distinct from the deterministic state s_t . I will compare the GRU with an LSTM and an explicit lifecycle updater, and test whether joint training improves on a frozen message encoder. Message-only and bounded-context baselines will establish whether sequence memory contributes beyond the current message (Figure 2).

Preliminary result and its boundary. Nested cross-validation selected a ModernBERT configuration with ten preceding messages. On the original 1,297-conversation test partition and its original labels, the selected system obtained scam $F_1 = 0.863 \pm 0.005$ across three seeds. This establishes a prototype baseline, not performance under the revised supervision. The revised label rule changes the target population, so retraining and fresh validation are required before a new headline result is reported. The Android integration demonstrates an implementation path; reliable early warning and protective impact have not been established by integration alone.

RQ-T: Can a behavioral sequence support timely, transferable warnings? I will first validate the revised annotation guide on a human-reviewed pilot, including all fourteen behaviors, legitimate money requests, uncertain cases, and behaviors missing from the vocabulary. The resulting eligibility decisions will govern training and testing. I will then compare model configurations with selection confined to training/validation data and use newly reserved evaluation conversations [27, 28]. Related posts, conversation fragments, and generated variants will remain in the same partition.

Evaluation will report class-specific precision/recall, macro- F_1 , calibration, and false alarms per conversation, alongside warning time. If d is the message position of the first validated exploitative demand and τ is a threshold chosen on validation data, define

$$\hat{t}_\tau = \min\{t : r_t \geq \tau\}, \quad L_\tau = d - \hat{t}_\tau. \quad (5)$$

Positive L_τ denotes a warning before the demand. I will report missed warnings separately, rather than omit them from average lead time; false alarms will also be measured in conversations without an exploitative demand.

ON-DEVICE DETECTION: TEXT PROTOTYPE AND PLANNED MODEL COMPARISONS

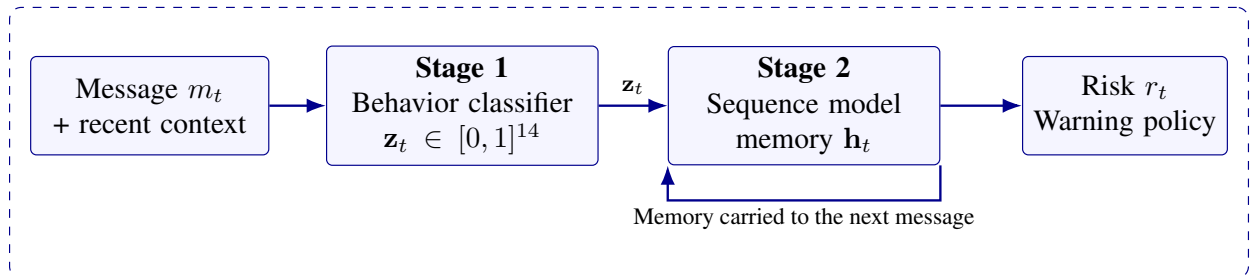


Figure 2: Two-stage detection design. The encoder–GRU text pipeline provides the prototype; model choice, warning thresholds, and device performance will be evaluated. Private text is processed locally, including a bounded context buffer.

Entire-source and family holdouts will test transfer within the families with sufficient examples. Adversarial experiments will insert plausible ordinary conversation before or between setup messages: such padding can push evidence out of bounded context or dilute a learned sequence. Paraphrases and delayed requests will test related failure modes. Comparisons will preserve the underlying scam evidence and measure both missed detection and additional warning delay. On-device replay will assess agreement with the research model, quantization effects, latency, peak memory, and energy on named hardware. Cross-application linkage and warning presentation will be evaluated as extensions, with explicit consent and human-subjects review before any user study.

6 Proposed Work: Credibility Artifacts in Conversational Scams

6.1 Problem and motivation

A conversational claim may be supported by a photograph, identity document, receipt, or investment screenshot. I use *credibility artifact* to mean media presented within an exchange as evidence for an identity, circumstance, capability, transaction, or opportunity. The term describes its use; it does not presume that the artifact is forged or malicious. Ordinary photographs and documents may serve the same function in legitimate exchanges.

The unresolved problem is whether these artifacts provide reliable additional evidence for scam detection, and whether that evidence arrives early enough to improve warnings. A text-only model may miss what a screenshot claims; an image classifier may miss why that claim matters to the request. Conversely, an apparently inconsistent or reused image may be benign, and a scammer can use genuine media. The proposed study will measure these cases before deciding how to incorporate artifact evidence into the detector. Its primary empirical scope is images and screenshots. Audio will be a conditional extension if usable original recordings become available; an audio-message icon in a screenshot is not an audio recording.

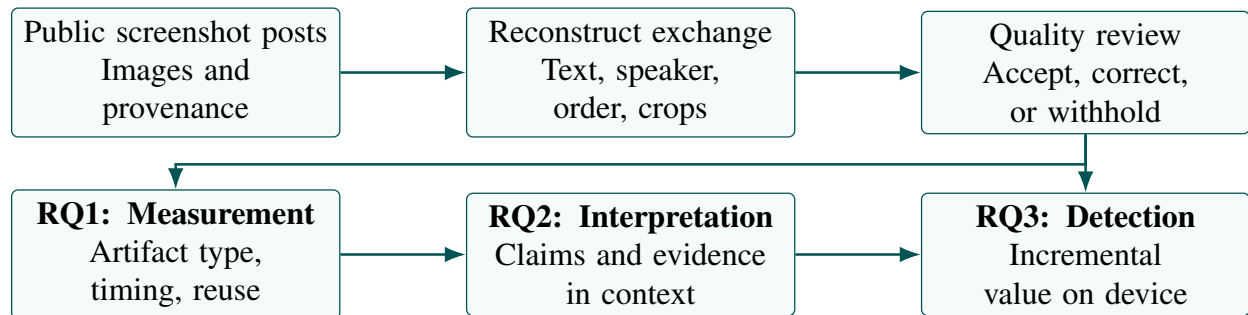


Figure 3: Proposed artifact study. Collection and extraction supply candidate evidence; quality review precedes measurement and detector evaluation. Artifact presence and source membership do not supply automatic scam labels.

RQ1: What artifacts appear, and how are they used over a conversation? Why this matters. A detector design needs evidence about which media are available, what claims they support, and when they appear. Without this measurement, a multimodal extension could focus on visually distinctive but rare or late-arriving content.

Approach. I will combine the existing screenshot collection with a continuing Reddit crawl and historical archive retrieval. The collection will retain post identifiers, gallery order, available media, and retrieval status. I will reconstruct messages and embedded artifact regions while retaining links to their source pixels. Text recognition, spatial layout, and a vision language model will be compared for transcription and speaker

attribution. Different extraction systems’ agreement will prioritize review, but a human sample of agreeing outputs is also necessary to detect shared mistakes.

Posts will be joined into a conversation only when continuity evidence supports the connection; matching an author or title is not sufficient. Screenshot overlap will be removed without deleting genuinely repeated messages. The review will distinguish the screenshot owner’s messages from those of the suspected scammer, retain uncertain roles, and record missing sections. I will audit extraction quality by app interface, image length, compression, and source before measuring artifact prevalence, first appearance, and within/across-conversation reuse. Exact and perceptual image matches will identify reuse candidates, followed by review. A match demonstrates shared content, not identity theft or common criminal control. Synthetic-image detectors may provide exploratory scores, but those scores will not be reported as the prevalence of AI-generated scam imagery without suitable ground truth and calibration.

Evaluation and expected contribution. The study will report message coverage, transcription error, speaker accuracy, artifact crop/link accuracy, and extraction yield before giving corpus-level measurements. A held-out audit will quantify selection and reconstruction limitations. The contribution will be a provenance-preserving measurement corpus and evidence about artifact use in observed exchanges, not a claim to represent all conversational fraud.

RQ2: Does an artifact support or conflict with the claims made in context? Why this matters. Recognizing a face or extracting a balance does not establish what the sender asserts. A uniformed photograph is inconsistent with an occupation claim only if the sender presents the depicted person as themselves and the context makes those claims incompatible. The study must distinguish contradiction from irrelevance, ambiguity, and missing evidence.

Approach and evaluation. I will annotate the artifact’s visible content, the claim explicitly associated with it, and the earlier messages needed to interpret that claim. Candidate relations will include support, contradiction, and insufficient information. Text-only, artifact-only, and joint models will be compared on reviewed examples, including genuine documents, benign image reuse, edited screenshots, and scams using apparently consistent media. Controlled mismatches will test specific capabilities but will be reported separately from naturally occurring cases. Evidence localization and claim-relation accuracy will accompany classification metrics. The expected contribution is an evaluated contextual-evidence representation, including its failure and abstention cases, rather than automatic verification of identity.

RQ3: Does artifact evidence improve warnings under mobile constraints? Hypothesis. When text alone leaves a claim ambiguous, an artifact’s content and its relation to prior messages may improve detection at an acceptable false-alarm rate. Additional model complexity is justified only if the measured benefit survives source separation and device constraints.

I will compare simple score combination with learned fusion. Let $\mathbf{e}_t$ represent the artifact evidence available at message t , and let u_t indicate whether an artifact is available. A candidate extension of Stage 2 is

$$\mathbf{h}_t^{\text{multi}} = g_\psi\left(\mathbf{h}_{t-1}^{\text{multi}}, [\mathbf{z}_t; \mathbf{e}_t; u_t]\right). \quad (6)$$

When no artifact is present, $u_t = 0$ and a masked representation supplies no invented evidence. This formulation does not require turning every image type into a new trigger. The experiment will determine whether the representation and fusion mechanism help beyond the existing behavioral sequence.

Evaluation will compare text alone, artifacts alone, and the combined system on the same eligible conversations, with component ablations. Related conversations and reused-image clusters will be separated across evaluation partitions to limit memorization. I will report recall at matched false-alarm rates, warning lead time, calibration, and robustness to cropping, compression, missing media, and benign media inserted to distract the detector. Device evaluation will include latency, energy, memory, and the cost of retaining context. The default deployment design will process private media locally. Any remote lookup would

require its own privacy and utility analysis; privacy does not follow from using an image hash.

6.2 Feasibility, risks, and completion criteria

A Reddit collection and screenshot-extraction prototype are already underway. Initial extraction runs demonstrate that structured text and artifact candidates can be recovered, while tall screenshots, speaker confusion, and tile overlap remain concrete challenges. These are feasibility observations, not a completed accuracy evaluation. Screenshot crops may be too small or degraded for reliable artifact analysis; the study will report that yield and can still characterize observable use and timing without claiming to determine authenticity.

The first milestone is a reviewed reconstruction sample and a frozen extraction contract. The second is a corpus measurement with documented selection and error rates. The third is a held-out comparison establishing when artifact evidence helps, harms, or adds no measurable value. A negative detection result remains informative if the evaluation identifies why the evidence is unavailable, unreliable, or redundant. Before release, personal identifiers will be reviewed and redacted where appropriate; data availability will reflect source permissions and human-subjects requirements rather than assume all collected media can be redistributed.

7 Research Plan and Expected Contributions

The remaining plan prioritizes reliable evidence before new model complexity. The first twelve months focus on the conversation-label pilot, reconstruction quality, text-model evaluation, and artifact measurement. The timetable below is provisional and depends on review capacity and the yield of usable media.

Period	Deliverable and decision point
Fall 2026	Prospectus defense; freeze and review the annotation guide; validate reconstruction on held-out screenshots; establish an eligible corpus and leakage-aware splits.
Spring 2027	Retrain/evaluate text models; measure warning timing, source/family transfer, and padding robustness; complete initial artifact measurements.
Summer 2027	Complete the contextual-claim reference set and compare interpretation methods; decide which artifact evidence merits detector integration.
Fall 2027	Evaluate fusion and device costs; investigate warning presentation and cross-app continuity within approved study scope. Prepare manuscripts.
Spring 2028	Resolve remaining evaluations, document limitations and artifacts, and complete dissertation writing and defense.

Table 3: Provisional milestones. Model expansion follows evidence and quality checks rather than a pre-determined accuracy or publication outcome.

The completed studies contribute methods and findings about application compliance and permission visibility. The ongoing study is expected to contribute a validated behavioral representation and an evaluation of sequential, on-device scam warnings. The proposed artifact study will add a contextual measurement resource and evidence about the value and limits of multimodal detection. Together, these contributions will connect mobile-security measurements to interventions whose benefits and limitations are explicitly tested.

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